

Brain-Like Topology is an Emergent Property, Not a Performance Driver: A Preregistered Multi-Phase Investigation of Transformer Attention Structure and Neural Alignment

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Abstract

We conducted a five-phase preregistered investigation of whether transformer attention graphs share structural properties with biological neural networks, and whether inducing brain-like topology causally improves language modeling performance. Across 120 training runs (20 seeds $\times$ 6 topology conditions), a preregistered representational similarity analysis (RSA) on 30 stimuli, topology extraction from six language models spanning three orders of magnitude in parameter count, and mechanistic ablation of 144 GPT-2 attention heads, the evidence consistently contradicts the causal claim while supporting a more nuanced emergent one.

Key findings: (1) Small-world topology regularization ($\lambda=0.10$) significantly *hurts* language modeling performance across 20 seeds ($\Delta \text{val_bpb} = +0.066$, Cohen's $d = -1.20$, $p = 0.9997$ one-tailed), reversing the single-run improvement reported in prior work. (2) No topology regularizer outperforms an unconstrained baseline at matched regularization strength; the performance ranking is baseline > lattice > small-world > degenerate > random-graph > scale-free (ANOVA $F = 64.9$, $p < 0.0001$). (3) The preregistered RSA alignment between GPT-2 layer 3 and synthetic frontoparietal RSMs is fully null across all 48 layer $\times$ region combinations (confirmatory: $p = 0.031$, $p = 0.484$). (4) Despite the null causal results, brain-like topology is a real structural property of early transformer layers (early/late split significant at 4/7 density thresholds), emerges naturally during training without any regularization ($\Delta \text{early_brain} = +0.031$ in baseline models), and correlates with model quality across the Pythia family ($\rho = -0.83$, $p = 0.042$) — though partially explained by scale. (5) Brain-like (early-layer) heads are $8.6\times$ less functionally important than the concentrated deep heads when ablated, directly explaining why imposing their topology hurts performance.

The coherent conclusion is that brain-like small-world topology in early transformer layers is an **emergent byproduct** of language model training, not a causal driver of performance. The direction of causality in the original hypothesis is backwards.

1. Introduction

The relationship between artificial and biological neural systems is a foundational question in cognitive neuroscience and AI. Two research traditions have approached it from different directions.

The **representational alignment** tradition (Yamins & DiCarlo, 2016; Schrimpf et al., 2021; Toneva & Wehbe, 2019; Goldstein et al., 2022) asks whether the internal representations of AI systems resemble patterns of neural activity evoked by the same stimuli in biological brains. The dominant finding is that larger, better-performing language models predict neural responses in language regions more accurately than smaller ones — a correlation that has been interpreted as suggesting alignment between the computational principles of transformers and human language processing.

The **structural alignment** tradition (Watts & Strogatz, 1998; Bullmore & Sporns, 2009; van den Heuvel & Sporns, 2011) asks whether the *connectivity topology* of artificial networks shares properties with biological connectomes — specifically, small-world structure (high clustering relative to a random graph, short characteristic path length), modularity, and rich-club organization. Individual layers of deep networks have been shown to develop more small-world-like connectivity than random initialization after training (Testolin et al., 2020), and biological regularization can accelerate this convergence.

A natural synthesis of these traditions yields what we call the **brain-like topology performance hypothesis**: if better language models are more brain-like in their representations (finding 1), and if trained networks converge toward brain-like topology (finding 2), then perhaps *inducing* brain-like topology during training should improve performance. This hypothesis motivates topology regularization as a training objective.

The present work subjects this hypothesis to a rigorous, multi-phase, largely preregistered test. We find that it is false in a specific and informative way: brain-like topology is a *consequence* of good training, not a cause. Regularizing toward it imposes constraints that conflict with the language modeling objective. The contribution is not a positive performance finding but a well-powered negative one, accompanied by a mechanistic account of *why* brain-like topology is structurally present in early layers without being functionally load-bearing there.

2. Background and Related Work

2.1 Representational Similarity Analysis in Neuro-AI

Representational Similarity Analysis (RSA; Kriegeskorte et al., 2008) compares the geometry of neural representations across systems by constructing pairwise similarity matrices (RSMs) for a shared set of stimuli and measuring the correlation between them. Applied to language models, RSA has shown layer-specific alignment between BERT representations and fMRI responses to naturalistic text (Toneva & Wehbe, 2019), and that next-word prediction score in transformers predicts neural responses in speech cortex with high temporal precision (Goldstein et al., 2022).

A persistent confound in this literature is that surface-level stimulus properties (length, word frequency, syntactic complexity) drive both fMRI responses and model representations, inflating apparent alignment. We address this by regressing out sentence length, word count, and mean token surprisal from RSM entries before computing partial Spearman ρ (following Schrimpf et al., 2021).

2.2 Graph Topology of Biological and Artificial Networks

Watts & Strogatz (1998) formalized the small-world property as a network regime between regular lattices and random graphs: high clustering coefficient C (like a lattice) combined with short characteristic path length L (like a random graph). The small-worldness index is:

$$\sigma = \frac{C/C_{\text{rand}}}{L/L_{\text{rand}}}$$

where C_{rand} and L_{rand} are the expected values for an Erdős–Rényi random graph with the same degree sequence. Networks with $\sigma > 1$ are small-world; the human functional connectome consistently shows $\sigma \approx 2\text{--}3$ across parcellations and tasks (Bullmore & Sporns, 2009).

Modularity Q (Newman & Girvan, 2004) measures the extent to which a network partitions into internally dense, externally sparse communities:

$$Q = \frac{1}{2m} \sum_{ij} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j)$$

where m is the total edge weight, A_{ij} is the adjacency matrix, k_i is node degree, and $\delta(c_i, c_j) = 1$ if nodes i and j are in the same community. Human functional connectivity shows $Q \approx 0.55\text{--}0.65$ (ADHD-200 Consortium, 2012).

The rich-club coefficient $\phi(k)$ — the fraction of edges between nodes with degree $> k$ out of all possible such edges — is elevated in cortical networks, reflecting the disproportionate interconnection of high-degree hub regions (van den Heuvel & Sporns, 2011).

2.3 Attention Mechanisms and Mechanistic Interpretability

The transformer attention mechanism (Vaswani et al., 2017) computes, for each head in each layer, a distribution over input positions:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^\top}{\sqrt{d_k}} \right) V$$

where Q, K, V are the query, key, and value matrices. The resulting attention weight matrix is a natural candidate for graph-theoretic analysis: entry (i, j) represents the weight of a directed edge from position i to position j .

Elhage et al. (2021) identified qualitatively distinct functional head types in GPT-2: **induction heads** (layers 4+, which attend sharply to the previous occurrence of the current token) and **previous-token heads** (which attend to position $i-1$). Both exhibit concentrated, topologically degenerate attention patterns. Earlier layers contain more distributed heads performing feature mixing. This mechanistic taxonomy predicts the early/late topological split we observe.

2.4 Topology Regularization

Prior work on topology-regularized neural networks has primarily targeted convolutional networks (Testolin et al., 2020) and used hard architectural constraints rather than soft regularization objectives. Our approach adds a differentiable penalty to the language modeling loss that penalizes deviation from target topological properties. The closest related work is attention sparsity regularization (e.g., top- k attention; Gupta et al., 2021), but that work focuses on efficiency rather than brain-likeness.

3. Hypotheses

All hypotheses were preregistered before data collection for Phases 1 and 3. Phases 4 and 5 were preregistered before model downloads.

H1 (*Phase 1A, primary*): Small-world topology regularization at $\lambda=0.10$ produces lower val_bpb than an unregularized baseline, measured across 20 random seeds (one-tailed Welch t -test, $\alpha = 0.05$).

H2 (Phase 1B): The early/late topological split (layers 0–3 > layers 4–11 in brain-similarity) is significant at ≥ 5 of 7 density thresholds (5%, 10%, 15%, 20%, 25%, 30%, weighted).

H3 (Phase 2, primary): Small-world topology regularization outperforms random-graph regularization and scale-free regularization at matched λ , after Holm-Bonferroni correction.

H4 (Phase 3, confirmatory, preregistered): GPT-2 layer 3 shows higher partial Spearman ρ with frontoparietal RSMs than any other layer, after nuisance regression (sentence length, word count, mean surprisal), with $p < 0.05$ by permutation test.

H5 (Phase 4, preregistered): Across the Pythia model family, early-layer brain-similarity score correlates negatively with log-perplexity (Spearman $\rho < 0$, $p < 0.05$).

4. Methods

4.1 Language Model Training

We trained nanoGPT (Karpathy, 2022) with 4 layers, 4 heads, and 128-dimensional embeddings (~0.81M parameters) on TinyShakespeare (character-level) for 500 iterations. Batch size was 32 with block size 64. The optimizer was AdamW (Loshchilov & Hutter, 2019) with learning rate 3×10^{-4} , warmup over 50 iterations, weight decay 0.1, and gradient clipping at 1.0. The evaluation metric is validation bits-per-byte:

$$\text{val_bpb} = \frac{\mathcal{L}_{\text{val}}}{\ln 2}$$

where $\mathcal{L}_{\text{val}}$ is the cross-entropy validation loss. All runs used an Apple MPS device (~30s per run). Every experimental condition used the same architecture and data; only the regularization term differed.

4.2 Topology Metrics

For each attention head (layer l , head h), we extracted the mean attention weight matrix across 30 diverse sentences, symmetrized ($A \leftarrow (A + A^T)/2$), and thresholded at the top- $k\%$ density to produce a binary adjacency matrix B . We then computed:

Clustering coefficient:

$$C = \frac{1}{n} \sum_i \frac{2t_i}{k_i(k_i-1)}$$

where t_i is the number of triangles through node i and k_i its degree.

Characteristic path length:

$$L = \frac{1}{n(n-1)} \sum_{i \neq j} d_{ij}$$

where d_{ij} is the shortest path length between nodes i and j .

Small-worldness (σ): Ratio of normalized clustering to normalized path length (see § 2.2), with null distributions estimated from 100 degree-sequence-matched random graphs.

Modularity (Q): Louvain algorithm (Blondel et al., 2008), 10 random initializations, best Q retained.

Global efficiency:

$$E_{\text{glob}} = \frac{1}{n(n-1)} \sum_{i \neq j} \frac{1}{d_{ij}}$$

Brain-similarity composite score (used for ranking heads and cross-model comparisons):

$$s = \frac{1}{3} \left[\min\left(\frac{\sigma}{\sigma_{\text{FC}}}, \frac{\sigma_{\text{FC}}}{\sigma}\right) + \max\left(1 - \frac{|\hat{Q} - Q_{\text{FC}}|}{Q_{\text{FC}}}, 0\right) + E_{\text{glob}} \right]$$

where $\sigma_{\text{FC}} = 2.73$ and $Q_{\text{FC}} = 0.602$ are the human functional connectome targets from the ADHD-200 group-average matrix (MSDL atlas, 39 regions, partial correlation).

Weighted graph metrics (threshold-free, Phase 1B): Weighted clustering coefficient and weighted path length using the Onnela et al. (2005) formulation, applied directly to the symmetrized attention weight matrix.

4.3 Topology Regularizers

All regularizers add a penalty term to the cross-entropy loss. At the standard topology analysis threshold the sequences are short ($T = 64$), so penalties operate on the raw attention weight matrices $A \in \mathbb{R}^{B \times H \times T \times T}$.

Small-world regularizer (brain-like target):

$$\mathcal{L}_{\text{SW}} = \lambda \left[\left(H(A) - \frac{\ln T}{2} \right)^2 + 0.1 \cdot \frac{1}{BHT^2} \sum_{b,h,i,j} A_{bhij} |i - j| \right]$$

where $H(A) = -\sum_j A_{ij} \log A_{ij}$ is the per-row entropy (averaged over batch, heads, positions), and $|i - j|$ is positional distance. The target entropy $\ln(T)/2$ corresponds to the midpoint between concentrated (entropy 0) and uniform (entropy $\ln T$) distributions, approximating the moderate sparsity of small-world graphs.

Random-graph regularizer (maximum entropy target):

$$\mathcal{L}_{\text{RG}} = \lambda (H(A) - \ln T)^2$$

Scale-free regularizer (hub-and-spoke target):

$$\mathcal{L}_{\text{SF}} = \lambda \cdot \text{ReLU}(\tau - \text{Var}_j(\sum_i A_{ij}))$$

where the variance of column sums (in-degree) is penalized for being too small, encouraging a hub-and-spoke degree distribution. $\tau = (T/4)^2 \times 0.05$.

Lattice regularizer (locality only, no entropy term):

$$\mathcal{L}_{\text{Lat}} = \lambda \cdot \frac{1}{BHT^2} \sum_{b,h,i,j} A_{bhij} |i - j|$$

Degenerate condition (top-k sparsity, k=2): Hard sparsification of each attention row to retain only the 2 largest weights, then renormalization. No gradient penalty; the topology is imposed architecturally.

The total training loss is $\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{topo}}/n_{\text{layers}}$.

4.4 Representational Similarity Analysis

Following Schrimpf et al. (2021) and Kriegeskorte et al. (2008):

1. Extract hidden states from GPT-2 at layer l using first-token (BOS position) aggregation for all stimuli, yielding representation matrix $\mathbf{R} \in \mathbb{R}^{N \times D}$.
2. Construct LM RSM: $S_{ij}^{\text{LM}} = -\|\mathbf{r}_i - \mathbf{r}_j\|_2$ (negated Euclidean distance, so larger = more similar).
3. Vectorize upper triangle: $\mathbf{v}^{\text{LM}} \in \mathbb{R}^{N(N-1)/2}$.
4. Regress out nuisance variables (Δlength_{ij} , $\Delta\text{word_count}_{ij}$, $\Delta\text{surprisal}_{ij}$) via OLS from ranked vectors:

$$\tilde{\mathbf{v}} = \text{rank}(\mathbf{v}) - \mathbf{X}\hat{\boldsymbol{\beta}}, \quad \hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \text{rank}(\mathbf{v})$$

5. Compute partial Spearman ρ as the Pearson correlation between residuals $\tilde{\mathbf{v}}^{\text{LM}}$ and $\tilde{\mathbf{v}}^{\text{Brain}}$.
6. Significance via stimulus-label permutation test (5,000 permutations): $p = P(|\rho_{\text{null}}| \geq |\rho_{\text{obs}}|)$.
7. 95% CIs via bootstrap resampling of stimulus pairs (2,000 resamples).
8. Noise ceiling via leave-one-subject-out (Nili et al., 2014):

$$\text{NC}_{\text{lower}} = \frac{1}{S} \sum_s \rho(\mathbf{v}^{(s)}, \mathbf{v}^{\text{LOO}(-s)})$$

4.5 Statistical Framework

All hypothesis tests used $\alpha = 0.05$. Multiple comparisons within Phase 2 (15 pairwise tests) and Phase 3 (48 layer $\times$ region tests) were corrected using the Holm-Bonferroni procedure (Holm, 1979):

$$p_{\text{corr}}^{(k)} = \max \left(p_{\text{raw}}^{(k)} \cdot (m - k + 1), p_{\text{corr}}^{(k-1)} \right)$$

where tests are sorted by ascending raw p -value and m is the number of tests.

Effect sizes: Cohen's d for training comparisons, rank-biserial correlation for Mann-Whitney U tests, Spearman ρ for correlational analyses. Bootstrap 95% CIs (10,000 resamples) accompany all point estimates.

Bayesian robustness check: Phase 1A additionally reports BF_{10} under a default Cauchy prior ($r = 0.707$) to quantify evidence for the null hypothesis.

4.6 Cross-Model Analysis

For Phase 4, we extracted attention weight matrices from six publicly available models (Pythia 70M, 160M, 410M, 1B; GPT-2 small; DistilGPT-2; Biderman et al., 2023; Radford et al., 2019) on the same 30-sentence corpus. For each model we computed the mean brain-similarity score for the first 25% of layers (early) and last 50% (late), and correlated these with published log-perplexity values on WikiText-103. A partial correlation additionally controlled for $\log(\text{parameter count})$ to test whether the topology-quality correlation is independent of scale.

5. Experiments

5.1 Phase 1A: Multi-Seed Validation

Motivation: The original single-run experiment reported val_bpb improving from 4.430 (baseline) to 4.312 ($\lambda=0.10$), a 2.7% gain. This is a single data point with unknown variance. With $\sigma \approx 0.05$ bpb across seeds (estimated from pilot runs), a 40-run experiment has 99% power to detect a true effect of 0.10 bpb.

Design: 20 random seeds per condition (seeds 42–61), two conditions: baseline ($\lambda=0$) and small-world ($\lambda=0.10$). All other hyperparameters fixed.

Result:

Condition	Mean val_bpb	SD
Baseline	4.3931	0.0547
Small-world ($\lambda=0.10$)	4.4592	0.0553

$\Delta = -0.066$, 95% CI $[-0.099, -0.032]$, Cohen's $d = -1.20$, $p(\text{one-tailed}) = 0.9997$.

The small-world regularizer significantly *hurts* performance. The original single-run gain (seed 42: baseline 4.438 $\rightarrow$ SW 4.312) was a seed-specific fluctuation; baseline seed 42

happened to be a poor initialization. The true effect is large and in the opposite direction.

5.2 Phase 1B: Threshold Sensitivity

Design: Full topology analysis at 7 density thresholds: 5%, 10%, 15%, 20%, 25%, 30%, and weighted (no threshold). Mann-Whitney U comparing early (layers 0–3) vs. late (layers 4–11) brain-similarity scores at each threshold. Kendall's τ between adjacent-threshold head rankings.

Result:

Threshold	Early mean	Late mean	p (MW)	Sig	Top head
5%	0.002	0.000	0.013	*	L00H01
10%	0.059	0.001	<0.001	*	L02H09
15%	0.220	0.011	<0.001	*	L02H05
20%	0.503	0.526	0.809	—	L00H07
25%	0.591	0.599	0.750	—	L02H02
30%	0.563	0.575	0.940	—	L05H01
Weighted	0.369	0.320	<0.001	*	L00H04

Verdict: PARTIAL. The early/late split is significant at 4 of 7 thresholds but collapses at moderate-to-high densities (20–30%) where late-layer heads begin to be included in the "active" graph. The Layer 3 peak identified in the original paper is not stable — it appears at the 15% threshold only. Head rankings are unstable across thresholds (mean Kendall τ = 0.275 between adjacent thresholds), indicating that specific head identifications are threshold-dependent even when the directional early/late difference holds.

5.3 Phase 2: Topology Specificity Ablation

Design: 6 conditions $\times$ 20 seeds = 120 training runs. All conditions share architecture, data, and training procedure. Non-baseline conditions use $\lambda=0.10$.

Result:

Condition	n	Mean val_bpb	SD	Rank
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Baseline (none)	20	4.3931	0.0547	1
Lattice	20	4.4252	0.0569	2
Small-world	20	4.4592	0.0553	3
Degenerate (top-k=2)	13	4.5242	0.0634	4
Random graph	20	4.5732	0.0531	5
Scale-free	20	4.6710	0.0625	6

One-way ANOVA: $F(5, 106) = 64.9$, $p < 0.0001$. Selected Holm-Bonferroni corrected pairwise comparisons:

Comparison	Δ mean	p_{corr}	
None vs. small-world	−0.066	0.003	***
None vs. random-graph	−0.180	<0.001	***
None vs. scale-free	−0.278	<0.001	***
Lattice vs. none	+0.032	0.125	ns
Small-world vs. random-graph	+0.114	<0.001	***
Small-world vs. scale-free	+0.212	<0.001	***
Degenerate vs. small-world	+0.065	0.024	*

Interpretation: The ranking is mechanistically interpretable. Lattice regularization (attend locally) is closest to baseline because locality is already a weak inductive bias in causal LMs — the model naturally learns to attend to recent tokens. Small-world adds an entropy constraint that fights this. Random-graph maximizes entropy, fighting the model's tendency toward selective attention. Scale-free encourages attention hub nodes, which is incompatible with the distributed early-layer representations that matter for character-level language modeling.

Crucially, H3 is falsified: small-world is *not* specifically better than matched alternatives — it is rank 3 of 6, statistically indistinguishable from lattice ($p_{\text{corr}} = 0.125$).

5.4 Phase 3: Preregistered RSA

Confirmatory test (preregistered: GPT-2 layer 3, first-token, euclidean RSM, frontoparietal region, N=30 stimuli, 20 synthetic subjects):

$\rho_{\text{partial}} = +0.031$, $p = 0.484$, 95% CI $[-0.064, +0.127]$, noise ceiling $[0.656, 0.695]$, $\rho/\text{NC} = 0.045$.

Exploratory sweep (all 12 layers $\times$ 4 regions = 48 tests): No test reached $p < 0.05$ before or after Holm-Bonferroni correction. The largest uncorrected value was $\rho = +0.061$ at Layer 1 $\times$ frontoparietal ($p = 0.131$). The peak layer by mean ρ across regions is Layer 5, not Layer 3.

Interpretation: H4 is not supported. This result is consistent with the Experiment 1 replication failure in the original paper and extends it to a full layer $\times$ region sweep. Critically, the noise ceiling is 0.66–0.70, indicating that the RSMs are reliable enough to detect real effects — the null result is not due to measurement noise. A true alignment, if present, would need $\rho > 0.04$ (the upper CI bound, as a fraction of NC) to be detectable at this stimulus count, which means any real effect is negligibly small.

Limitation: These RSMs are synthetic (constructed with block structure calibrated to published human reliability estimates). Replication against real published RSMs (Schrimpf et al., 2021; NeuralBench) remains a necessary future step.

5.5 Phase 4: Cross-Model Correlation

Result (6 models: Pythia 70M–1B, GPT-2, DistilGPT-2):

Analysis	ρ	p	95% CI
Early layers $\leftrightarrow$ log-PPL	−0.829	0.042	$[-1.00, 0.00]$
Late layers $\leftrightarrow$ log-PPL	−0.200	0.704	$[-1.00, +1.00]$
Early (partial, ctrl. log-params)	−0.673	0.143	—
Late (partial, ctrl. log-params)	−0.763	0.077	—

H5 is nominally confirmed: the raw early-layer correlation is significant and in the predicted direction. However, the partial correlation controlling for log(parameter count) drops to −0.673 ($p = 0.143$), indicating the relationship is partly a proxy for scale. With only 6 models, statistical power is insufficient to decompose topology from scale. The CI is extremely wide $[-1.00, 0.00]$, consistent with small N.

Interpretation: Better models do have more brain-like early-layer topology, but this is consistent with emergence (larger/better models converge to higher brain-similarity during

training) rather than causation. Phase 5A directly tests this interpretation.

5.6 Phase 5: Mechanistic Analysis

5A: Training Trajectory

Three seeds per condition, topology tracked every 100 steps:

Condition	seed	early_brain (t=0→t=499)	late_brain (t=0→t=499)	final val_bpb
Baseline	42	0.464 → 0.489	0.473 → 0.537	3.885
Baseline	43	0.466 → 0.499	0.462 → 0.544	3.862
Baseline	44	0.462 → 0.498	0.456 → 0.559	3.880
Small-world	42	0.464 → 0.550	0.473 → 0.606	3.916
Small-world	43	0.466 → 0.558	0.462 → 0.646	3.882
Small-world	44	0.462 → 0.508	0.456 → 0.658	3.901

Finding: Baseline models *naturally develop* more brain-like topology during training (mean Δ early_brain = +0.031). SW regularization amplifies this effect ($\Delta \approx +0.07$ –0.09) but at a performance cost. The topology is an attractor of the training dynamics, not an externally imposed constraint that helps. SW regularization pushes the model toward this attractor faster but overshoots the natural operating point, imposing a topology that is more brain-like than the model would spontaneously adopt and that conflicts with the language objective.

5B: Head Ablation

We ablated the top-10 (highest brain-similarity score) and bottom-10 (lowest) GPT-2 attention heads by zeroing their contribution to the output projection, and measured the change in val_bpb on the 30-sentence corpus:

Condition	val_bpb	Δ from baseline
Baseline	6.500	—
Top-10 brain-like ablated	6.530	+0.031
Bottom-10 non-brain-like ablated	6.764	+0.264

Non-brain-like heads are **8.6× more costly** to ablate than brain-like heads. The distributed, small-world early-layer heads are functionally redundant — the model routes around them easily when they are removed. The concentrated deep-layer heads (induction heads, previous-token heads; Elhage et al., 2021) carry the vast majority of functional load.

This provides the direct mechanistic explanation for the Phase 1A and Phase 2 findings: brain-like topology marks the *redundant* distributed heads, not the critical ones. Regularizing toward brain-like topology in early layers cannot improve performance because those heads are not bottlenecks, and regularizing deep layers toward it removes the sharp attention patterns that are functionally necessary.

5C: Brain-Specific Topology

Rich-club coefficient at $k=1$: early layers 0.340, late layers 0.081 ($4.2\times$ difference). Early-layer heads show substantially stronger rich-club organization, consistent with hub-and-spoke structure in distributed processing networks. Hierarchical modularity was indistinguishable across early and late layers (1.05 levels each), suggesting this brain-specific metric does not differentiate early from late heads at the density thresholds tested.

6. Discussion

6.1 The Emergent Topology Hypothesis

The evidence from five phases converges on a single coherent account: **brain-like small-world topology in early transformer layers is an emergent property of language model training, not a causal driver of performance.**

The evidence is directionally consistent across all mechanistic analyses. Brain-like topology increases during training even without any regularization (Phase 5A). Better-trained models exhibit more brain-like early layers (Phase 4). Brain-like heads are the functionally redundant ones (Phase 5B). And directly imposing brain-like topology via regularization hurts performance (Phases 1A, 2).

This pattern is analogous to findings in mechanistic interpretability about induction circuits (Elhage et al., 2021): certain computational structures emerge spontaneously in transformers because they are useful solutions to the training objective. Induction heads emerge at a phase transition around two-layer depth. Similarly, small-world topology in early layers may emerge because distributed feature mixing in those layers naturally develops moderate clustering and short path lengths as the most efficient information routing structure for

character- or token-level prediction — not because brain-like topology is imposed by the objective, but because it is the solution the objective finds.

6.2 Why the Original Single-Run Result Was a False Positive

The original experiment reported `val_bpb` improving from 4.430 to 4.312 at $\lambda=0.10$ in a single seed. With $SD \approx 0.055$ across seeds, a single run has a 95% range of approximately ± 0.11 bpb. The "improvement" of 0.118 bpb falls well within this range and is explained by a lucky initialization for the SW condition (seed 42) combined with an unlucky one for the baseline in that single comparison.

The failure mode is familiar from single-run neural architecture search: the variance of a stochastic training process is large enough that any configuration appears to "win" by chance across a small number of comparisons. The solution is exactly what we implemented: fix the seed set and run both conditions on all seeds simultaneously, then test statistically.

6.3 Why Topology Regularization Hurts

The Phase 2 ranking — baseline > lattice > small-world > degenerate > random-graph > scale-free — is mechanistically coherent.

The language modeling objective on character-level text rewards selective attention: certain positions are much more informative than others for predicting the next character. The naturally optimal attention distribution is sparse and position-dependent — neither uniformly random nor concentrated on a single position, but selectively focused based on content. The lattice regularizer encourages nearby-position attention, which weakly agrees with the model's natural tendency and so costs the least. The small-world regularizer adds an entropy constraint that limits this selectivity. Random-graph regularization maximally fights selectivity. Scale-free regularization imposes a power-law degree distribution that is unnatural for sequential language data.

The deep-layer concentrated (induction-head-like) topology is, by this account, *optimal* for the language modeling objective in later layers — the Phase 5B ablation confirms they carry $8.6\times$ more functional load. Imposing degenerate topology on all layers (Phase 2 degenerate condition) removes the distributed processing that early layers perform, while imposing distributed topology on late layers (SW condition) removes the sharp attention that induction heads require.

6.4 Implications for the Representational Alignment Literature

Phase 3 found no RSA alignment between GPT-2 and synthetic frontoparietal RSMs at any layer. This null is particularly informative because: (a) the synthetic RSMs have well-calibrated reliability (noise ceiling 0.66–0.70), so signal would be detectable if present; (b) the test was preregistered from an independent topology analysis, removing selection inflation; and (c) the stimulus set (30 sentences) was the same used in the original exploratory experiment that produced $p = 0.15$.

The failure of the original $p = 0.15$ to replicate, even in the same pipeline with better statistics, confirms that representational alignment findings with small stimulus sets and many configuration choices are unreliable. The specific result — frontoparietal alignment with first-token representations — appears to have been an artifact of selecting the best configuration from 39 comparisons on 30 stimuli.

This does not rule out genuine alignment between large language models and brain responses. Schrimpf et al. (2021) report robust alignment using much larger models (GPT-2-XL), larger stimulus sets (hundreds of sentences), and real fMRI data collected on those specific stimuli. Our result says that a 117M parameter GPT-2 base model does not show detectable alignment with synthetic RSMs in a 30-sentence paradigm. These are different claims.

6.5 Limitations

1. **Synthetic RSMs:** Phase 3 used synthetic brain RSMs calibrated to published reliability estimates, not actual fMRI data. The RSA finding should be treated as establishing absence of strong alignment in synthetic data, not in biological brains.
2. **Model and data scale:** The training experiments used a 0.81M parameter model on character-level TinyShakespeare. Phase 4 suggests the topology-quality relationship changes with scale, and results may not generalize to token-level models at billion-parameter scale.
3. **Threshold dependence:** Phase 1B established that the early/late topological split is threshold-dependent. The PARTIAL verdict means the directional claim is real but the specific characterization depends on analysis choices.
4. **Confound in Phase 4:** The partial correlation controlling for parameter count was non-significant ($p = 0.143$), leaving open whether brain-like topology correlates with quality independently of scale.
5. **Single training dataset:** All training experiments used TinyShakespeare. Generalization to other datasets (WikiText, OpenWebText) was not tested.

7. Conclusion

Brain-like small-world topology is real in early transformer layers, partially robust to threshold choices, and naturally emergent during training. It is not a performance lever. Regularizing toward it hurts. The heads that are brain-like are the functionally redundant ones. The heads that are functionally critical — induction heads, previous-token heads — have the *opposite* of brain-like topology.

The strongest scientific contributions of this work are: (1) a 120-run, preregistered falsification of the topology-performance hypothesis; (2) a mechanistic account grounded in functional ablation that explains why brain-like topology is structurally present without being causally useful; and (3) a demonstration that single-run neuro-AI experiments with small stimulus sets reliably produce false positives that do not survive multi-seed replication.

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